

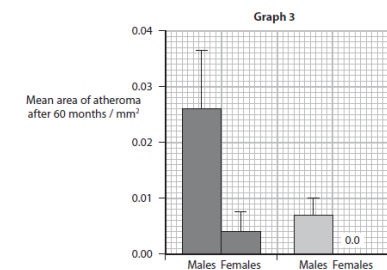
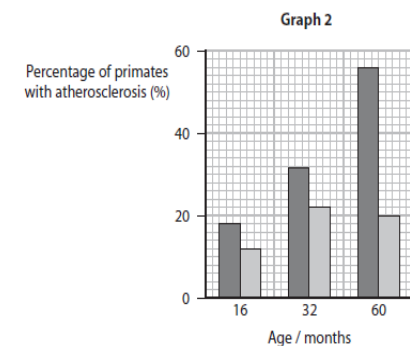
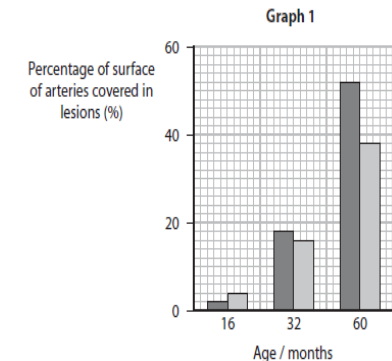
Question number	Answer	Additional guidance	Mark
8(a)	An answer that includes the following points: • at least one carbon-carbon double bond drawn (1) • rest of diagram correct (remaining CC single bonds and {Hs added /Lewis structures}) (1)	DO NOT ACCEPT double bond between C1 and C2 ecf from double bond between C1 and C2 ecf from saturated fatty acid	(2)

Question number	Answer	Additional guidance	Mark
8(b)(i)	• 1.1 / 1.14 / 1.136 (1)		(1)

Question number	Answer	Additional guidance	Mark
8(b)(ii)	• 1.5 : 1 / 1.45 : 1 (1)		(1)

Question number	Answer	Additional guidance	Mark
8(b)(iii)	• 0.022 (1) • 22 000 / 2.2×10^4 (1)	Bald answer of 22000 / 2.2×10^4 = 2 marks Bald answer of 22 of any magnitude or incorrect standard form format = 1 mark	(2)

Question number	Answer
*8(b)(iv)	Graph 1 : surface area of arteries Conclusions : - arteries have lesions with both lipids - percentage of surface covered in lesions increases with time for <u>both</u> sat and unsat lipids (overall) increase is greater (over time) with sat lipid / converse - type of lipid in diet does influence the percentage of arteries covered with lesions - data may not be {valid as no experimental details / repeatable as no error bars} (only once) Graph 2 : % atherosclerosis Conclusions : - atherosclerosis occurs with both lipids - atherosclerosis is higher with sat lipids in all three ages /converse - percentage increases with age for sat lipids - age has a less clear effect with unsat lipids - data may not be {valid as no experimental details / repeatable as no error bars / significant as no stats test} (only once) Graph 3 : area of atheroma Conclusions : - atheromas are larger in <u>both</u> males and females with sat lipids than unsat lipids - males have atheromas of greater area than females whichever lipid they are given - sex is an influence on area as well - data likely to be significant because error bars do not overlap - male data more variable



			Additional guidance
Level 0	0	No awardable content	
Level 1	1-2	Demonstrates isolated elements of biological knowledge and understanding to the given context with generalised comments made. Vague statements related to consequences are made with limited linkage to a range of scientific ideas, processes, techniques and procedures. The discussion will contain basic information with some attempt made to link knowledge and understanding to the given context.	Simple descriptions of data 1 mark = description of data in one graph 2 marks = description of data in two graphs
Level 2	3-4	Demonstrates adequate knowledge and understanding by selecting and applying some relevant biological facts / concepts. Consequences are discussed which are occasionally supported through linkage to a range of scientific ideas, processes, techniques and procedures. The discussion shows some linkages and lines of scientific reasoning with some structure.	Extended links made 3 marks = description of data in three graphs 4 marks = conclusion for one graph
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting and applying relevant biological facts / concepts. Consequences are discussed which supported throughout by sustained linkage to a range of scientific ideas, processes, techniques and procedures. The discussion shows a well-developed and sustained line of scientific reasoning which is clear and logically structured.	Understanding of graphs used to comment on effect of dietary supplements 5 marks = conclusion for two graphs 6 marks = conclusion for three graphs

Question number	Answer	Additional guidance	Mark
8(b)(v)	An answer that includes three of the following points: Unethical : • may cause {pain / harm / discomfort} / it is cruel (1) • animals may die (1) • animals should not be kept in captivity / being kept in captivity caused stress (1) • animals unable to consent (1) Ethical : • it is more ethical than experimenting on humans (1) • primates are (more) similar to humans (than other animals) (1) • might improve {human / animal} health (in future) (1)	ACCEPT may cause health issues ACCEPT against their will they have no say in the matter have not given their permission animals have their own rights ACCEPT humans are more important {should not harm / kill} humans	(3)